



Department of Computer Engineering
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Subject: DBMS Lab

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Experiment No. 5

Aim: Implement various Join operations.

Exercise 1: Demonstrate the application of INNER JOIN, NATURAL JOIN, and OUTER JOIN on the tables available in your system.

```
mysql> SELECT
->
->
->      c.fname,
->      c.lname,
->      e.location
-> FROM candidate c
->
-> RIGHT JOIN elections e ON c.eid = e.eid;
```

fname	lname	location
Amit	Patel	Thane East
Sita	Rao	Thane East
Aditi	Sharma	Thane
Rahul	Verma	Thane
choi	mina	india
choi	rina	india, mumbai
NULL	NULL	india, mumbai



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```
mysql> SELECT
->     c.fname,
->     c.lname,
->     p.party_name
-> FROM candidate c
-> LEFT JOIN party_member p ON c.candidate_id = p.candidate_id;
```

fname	lname	party_name
Aditi	Sharma	Progressive Party
Rahul	Verma	NULL
choi	mina	NULL
choi	rina	NULL
Amit	Patel	Blue Party
Sita	Rao	NULL
John	Doe	Green Party

7 rows in set (0.00 sec)

```
mysql> SELECT
->     fname,
->     lname,
->     party_name
-> FROM candidate
-> NATURAL JOIN party_member;
```

fname	lname	party_name
Aditi	Sharma	Progressive Party
Amit	Patel	Blue Party
John	Doe	Green Party

3 rows in set (0.01 sec)

```
mysql> SELECT
->     fname,
->     lname,
->     party_name
-> FROM candidate
-> NATURAL JOIN party_member;
```

fname	lname	party_name
Aditi	Sharma	Progressive Party
Amit	Patel	Blue Party
John	Doe	Green Party

3 rows in set (0.01 sec)



Exercise 2 :

Write a query to display student names along with their department names using INNER JOIN.

```
mysql> SELECT Student.StudentName, Department.DeptName
-> FROM Student
-> INNER JOIN Department ON Student.DeptID = Department.DeptID;
+-----+-----+
| StudentName | DeptName |
+-----+-----+
| Srushti     | CS       |
| Srushti     | Math     |
| Lavanya     | IT       |
+-----+-----+
3 rows in set (0.00 sec)
```

Write a query to display all students and their department names using LEFT OUTER JOIN.

```
mysql> Select S.StudentId, S.StudentName, S.DeptId, D.DeptName from Student S left join Department D on S.DeptId=D.DeptId;
+-----+-----+-----+-----+
| StudentId | StudentName | DeptId | DeptName |
+-----+-----+-----+-----+
| 101 | Srushti | 201 | CS |
| 102 | Srushti | 202 | Math |
| 103 | Lavanya | 203 | IT |
| 104 | Krishna | 208 | NULL |
| 105 | Pranav | 209 | NULL |
| 106 | Atharva | 210 | NULL |
+-----+-----+-----+-----+
6 rows in set (0.02 sec)
```

Write a query to display all departments and their students using RIGHT OUTER JOIN.

```
mysql> Select S.StudentId, S.StudentName, S.DeptId, D.DeptName from Student S right join Department D on S.DeptId=D.DeptId;
+-----+-----+-----+-----+
| StudentId | StudentName | DeptId | DeptName |
+-----+-----+-----+-----+
| 101 | Srushti | 201 | CS |
| 102 | Srushti | 202 | Math |
| 103 | Lavanya | 203 | IT |
| NULL | NULL | NULL | COMPS |
| NULL | NULL | NULL | COMPS |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```



Write a query using NATURAL JOIN to display matching student and department records

```
mysql> SELECT StudentName, DeptName
-> FROM Student
-> NATURAL JOIN Department;
+-----+-----+
| StudentName | DeptName |
+-----+-----+
| Srushti     | CS       |
| Srushti     | Math     |
| Lavanya     | IT       |
+-----+-----+
3 rows in set (0.00 sec)
```

Conclusion: This experiment helped in understanding how SQL joins are used to combine data from multiple related tables. By implementing different types of joins, we observed how matching and non-matching records are retrieved based on relationships between tables. It demonstrated the importance of primary and foreign keys in maintaining data integrity. Overall, joins play a vital role in efficient data retrieval in relational databases.